Commissioned Module/Pathway/Taxon Review — Rubredoxin Electron Transfer in *Pseudomonas putida* KT2440

Module/Bucket: UniPathway UPA00191 (`unipathway:UPA00191`) **Target taxon:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Candidate genes from local metadata:** 1 — `rubA` (PP_5315 | Q88C68 | Rubredoxin) **Curation verdict (headline):** Electron-carrier step **COVERED**; terminal alkane-monooxygenase step **GAP / not_expected_in_target_taxon**; UPA00191 as an alkane-degradation pathway is **NOT satisfiable**; RubA's alkane-hydroxylase coupling is a **likely over-propagated annotation**.

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1. Executive Summary

The UniPathway UPA00191 module for *Pseudomonas putida* KT2440 rests on a single candidate gene, `rubA` (PP_5315, UniProtKB:Q88C68), an iron-binding rubredoxin annotated as an electron-transfer component of a hydrocarbon-hydroxylating system. This review evaluated whether the module is satisfiable in KT2440 and whether the alkane-oxidation framing of RubA holds up against direct sequence, genomic-context, and literature evidence. The short answer: the **electron-carrier machinery is genuinely present, but the pathway it is supposed to feed is not.**

RubA is a *bona fide* single-domain rubredoxin (55 aa) with two canonical CxxCG iron-binding motifs, sitting immediately adjacent to a FAD-dependent rubredoxin–NAD(+) reductase, AlkT (PP_5314, K05297 / EC 1.18.1.1). KT2440 additionally carries a second, fused reductase–rubredoxin protein (PP_5371). So the reductase → rubredoxin electron-relay hardware — NADH → AlkT → RubA — is encoded and structurally intact. That part of the module is **covered**.

However, the terminal enzyme that this relay is meant to power — a membrane non-heme diiron **alkane 1-monooxygenase (AlkB, EC 1.14.15.3)** — is **absent** from the KT2440 proteome. The only gene named "alkB" in KT2440 (PP_3400) is a **homonym**: it is the DNA-repair α-ketoglutarate/Fe(II)-dependent dioxygenase AlkB (KEGG K03919), an entirely different enzyme.

The prototypical alkane-degradation system (`alkBFGHJKL`/`alkST`) resides on the OCT plasmid of *P. putida* GPo1 / *P. oleovorans*, which KT2440 does not carry. KT2440 is not an alkane-degrader. Consequently, UPA00191 read as "alkane degradation via alkane 1-monooxygenase" is **not satisfiable** in this organism, and RubA's `GO:0043448 alkane catabolic process` / UPA00191 assignments (all IEA, propagated by pathway inference rather than experiment) are **over-propagated**. Sequence typing reinforces this: RubA is a conserved single-domain **AlkG1-type housekeeping rubredoxin** (90% identical to a conserved *Ectopseudomonas (Pseudomonas) oleovorans* rubredoxin, only ~54–56% to the two-domain alkane-transfer AlkG/Rd2), most plausibly a general NAD(P)H:rubredoxin electron carrier — not a dedicated alkane-hydroxylase partner.

Curation bottom line: Keep the *rubredoxin electron-transfer step* as **covered** (RubA + AlkT present), but mark the overall alkane-degradation module as **gap / not_expected_in_target_taxon** at the terminal monooxygenase step, flag RubA's alkane-specific role as **candidate_uncertain (likely over-annotation)**, and treat the whole UPA00191 module as **module_needs_revision** for this organism. The generic module boundary (rubredoxin = "alkane degradation") is **wrong for KT2440**.

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2. Target-Organism Pathway Definition

What UPA00191 is meant to include. UniPathway UPA00191 is defined around the rubredoxin-dependent electron-transfer step that supplies reducing equivalents to a hydrocarbon (alkane) hydroxylating system. In the canonical, best-characterized version of this chemistry — the *P. putida* GPo1 / *P. oleovorans* OCT-plasmid Alk system — the electron flow is:

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NADH → AlkT (rubredoxin reductase, FAD) → AlkG (rubredoxin, Fe) → AlkB (alkane 1-monooxygenase)
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RubA (with AlkT) corresponds to the **reductase/carrier** components of that chain. The module scope as commissioned is narrow: "rubredoxin electron transfer" for "hydrocarbon hydroxylation." This is the canonical two-electron delivery system for AlkB activation (Williams et al., 2022, PMID: 34990970: *"Electrons are typically supplied by NADH via a rubredoxin reductase (AlkT) to a rubredoxin (AlkG) to AlkB"*).

What must be kept separate for KT2440. For curation purposes it is essential to distinguish three things that share names or EC-space but are biologically distinct in KT2440:

1. **Alkane 1-monooxygenase AlkB (EC 1.14.15.3)** — a ~400-aa integral-membrane non-heme diiron hydroxylase. This is the pathway's terminal enzyme. It is **not present** in KT2440.
2. **DNA-repair AlkB (K03919, EC 1.14.11.-)** — a ~215-aa cytoplasmic α-ketoglutarate/Fe(II) dioxygenase that oxidatively demethylates alkylated DNA. This *is* present (PP_3400) but is unrelated to hydrocarbon catabolism. It must not be counted toward UPA00191.
3. **General rubredoxin/reductase redox housekeeping** — RubA + AlkT + PP_5371 function as a NAD(P)H:rubredoxin electron-relay module that in KT2440 is associated with **fatty-acid degradation (KEGG ppu00071)** and general redox/oxidative-stress physiology, *not* a dedicated alkane pathway.

Alternate names / database definitions. UPA00191 = "alkane degradation" (UniPathway); GO:0043448 = "alkane catabolic process." RubA = "Rubredoxin," locus PP_5315, UniProtKB Q88C68. In the alkane literature the functional rubredoxins are called **Rd2 / AlkG (AlkG2-type)** and the non-functional conserved ones **AlkG1-type**. The adjacent reductase AlkT (PP_5314) is variously "rubredoxin reductase," "rubredoxin–NAD(+) reductase" (EC 1.18.1.1), KEGG K05297. Curators should be alert that "AlkB" and "AlkG/AlkT" naming is heavily overloaded across databases.

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3. Expected Step Model

The minimal expected-step model for a satisfiable UPA00191 alkane-hydroxylation module, and its status in KT2440:

Step	Enzyme / role	EC / KO	Expected gene	KT2440 status
S1	Rubredoxin–NAD(+) reductase (FAD); NADH → rubredoxin	EC 1.18.1.1 / K05297	alkT	**Present** — PP_5314 (Q88C69); also fused PP_5371 (Q88C12)
S2	Rubredoxin electron carrier; reductase → hydroxylase	GO:0009055; Fe binding	rubA / alkG	**Present but reclassified** — PP_5315 (Q88C68), AlkG1-type housekeeping rubredoxin
S3	Alkane 1-monooxygenase; hydroxylates alkane	EC 1.14.15.3	alkB	**Absent** — no true membrane diiron AlkB; PP_3400 is a homonymous DNA-repair dioxygenase (K03919)

Satisfiability conclusion: Steps S1 and S2 (the electron-transfer components that the module nominally covers) are encoded. Step S3, the oxygen-activating catalytic terminus that gives the pathway biological meaning, is **missing**. A pathway that supplies electrons to an enzyme that does not exist in the organism cannot be considered satisfiable. UPA00191 as an *alkane-degradation* module is therefore **not satisfiable** in KT2440; the rubredoxin/reductase genes are best modeled as a general electron-transfer couple rather than as members of an alkane-catabolic pathway.

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4. Candidate Genes and Evidence

4.1 RubA — PP_5315 (Q88C68) — Rubredoxin *[primary candidate]*

Likely role: Genuine single-domain rubredoxin electron carrier; small Fe(3+) redox protein. Evidence is strong that it *is* a rubredoxin; evidence that it feeds an alkane monooxygenase is weak/over-propagated.

Evidence type and detail (Findings F001, F004): - UniProt Q88C68 describes a 55-aa rubredoxin with an Fe(3+) cofactor and **two canonical iron-binding CxxCG motifs** (C6IVCG10 and C38PDCG42). - Domain signatures: InterPro **IPR024922** (rubredoxin) and **IPR018527** (rubredoxin, iron binding site); Pfam **PF00301**. - Architecture is a **single rubredoxin domain (~55 aa)**. This is the decisive discriminator: functional alkane-transfer rubredoxins (GPo1

AlkG/Rubredoxin-2, P00272, 173 aa; *P. putida* AlkG, Q9WWW4, 175 aa) are **two-domain ~173–175-aa** proteins whose C-terminal metal domain carries the signature `WICITCGHIYDEAL...CCPDCGATKEDY`. RubA lacks this second domain. - Anchored pairwise identity: **90%** to a conserved single-domain rubredoxin of the alkane-degrader *Ectopseudomonas (Pseudomonas) oleovorans* (A0A061CWA0, 55 aa), but only **~54–56%** to the alkane-competent two-domain donors (P00272; Q9WWW4). Even the alkanotroph *P. oleovorans* keeps this single-domain rubredoxin as a **separate gene** from its functional AlkG.

Curation-relevant caveats (Finding F003): - All of RubA's pathway/GO annotations are **electronic (IEA)**: `GO:0043448 alkane catabolic process` (IEA:UniProtKB-UniPathway), `GO:0009055 electron transfer activity` (IEA:InterPro), and UPA00191 itself is assigned by **pathway propagation, not experiment**. - Functional typing (van Beilen 2002, PMID: 11872724) places RubA in the **AlkG1 housekeeping class** — conserved across gram-positive and gram-negative bacteria including non-alkane-degraders, and demonstrated *not* to complement alkane hydroxylation. The alkane-catabolic framing is therefore an over-propagation. - **Status: covered** for a generic rubredoxin electron-transfer role; **candidate_uncertain** for the alkane-specific role.

4.2 AlkT — PP_5314 (Q88C69) — Rubredoxin–NAD(+) reductase *[adjacent, not in candidate list]*

Likely role: FAD-dependent rubredoxin reductase (382 aa); oxidizes NADH and reduces RubA. KEGG **K05297 / EC 1.18.1.1**, KEGG pathway ppu00071. Immediately adjacent to RubA (PP_5315), forming a discrete **two-gene electron-transfer couple**. The other genomic neighbor, PP_5313, is HU-alpha (a nucleoid protein, unrelated), confirming this is a minimal RubA+AlkT unit and **not** part of a larger `alk` operon (Finding F001). Evidence: homology-based; annotation appears correct.

4.3 PP_5371 (Q88C12) — fused reductase–rubredoxin *[additional, not in candidate list]*

Likely role: A second FAD + Fe **fused** rubredoxin-reductase protein (461 aa; FAD-dependent oxidoreductase domain + rubredoxin domain), KEGG K05297, ppu00071. Provides redundant NAD(P)H:rubredoxin reductase capacity and carries the *same* over-propagated "hydrocarbon hydroxylating system → alkane 1-monooxygenase" text. Represents **paralog ambiguity**: a second, self-contained NADH→rubredoxin relay, reinforcing that KT2440 maintains general rubredoxin-based electron transfer independent of any alkane pathway.

4.4 PP_3400 (Q88HF9) — "alkB" homonym *[a trap, not a pathway member]*

Likely role: DNA oxidative demethylase / DNA-repair AlkB. UniProt gene name is "alkB," 215 aa, Fe(2+) cofactor, keyword "Dioxygenase," protein existence "Predicted." KEGG assigns ppu:PP_3400 to **K03919** ("DNA oxidative demethylase AlkB," α-ketoglutarate/Fe(II)-dependent dioxygenase, EC 1.14.11.-). This is **not** the membrane non-heme diiron alkane 1-monooxygenase (EC 1.14.15.3). Any automated satisfiability routine that matches on the gene name "alkB" will **falsely** conclude the pathway is complete; PP_3400 must be explicitly excluded (Finding F002).

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5. Gaps, Ambiguities, and Likely Over-Annotations

1. **Terminal-step GAP (decisive).** No true alkane 1-monooxygenase (EC 1.14.15.3) exists in UP000000556. The prototypic system `alkBFGHJKL` + `alkST` is OCT-plasmid-borne in *P. putida* GPo1 / *P. oleovorans*; KT2440 does not carry it and does not grow on n-alkanes. This is the single most important curation fact: **the pathway lacks its catalytic terminus.**
2. **Homonym ambiguity (AlkB).** The DNA-repair AlkB (PP_3400, K03919) and the alkane hydroxylase AlkB (EC 1.14.15.3) share a name but nothing else — same three-letter name, different biochemistry. This is a classic source of false module satisfaction and must be flagged.
3. **Over-propagated alkane annotation on RubA (and PP_5371).** UniProt FUNCTION / UPA00191 / GO:0043448 are all IEA propagations from the GPo1 Alk paradigm, not experimentally supported for KT2440. Single- vs two-domain typing (single-domain, AlkG1-class, 90% to a housekeeping *P. oleovorans* rubredoxin, ~55% to two-domain AlkG2/Rd2) shows RubA is a housekeeping electron carrier; the alkane-specific function does not transfer to KT2440.
4. **Paralog redundancy in the electron-relay layer.** Two reductase routes to rubredoxin (standalone AlkT PP_5314 + fused PP_5371, both K05297) suggest a general/redundant electron-transfer role rather than a single dedicated alkane pathway. KEGG associates them with fatty-acid degradation (ppu00071), not alkane degradation.
5. **Broad EC/GO mappings.** `GO:0009055 (electron transfer activity)` and `GO:0005506 (iron ion binding)` are correct but generic; `GO:0043448 (alkane catabolic process)` is the problematic over-specific term.

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6. Module and GO-Curation Recommendations

Module step	Recommended status	Rationale
Rubredoxin reductase (AlkT, S1)	**covered**	PP_5314 (+PP_5371), K05297 / EC 1.18.1.1 present, adjacent to RubA
Rubredoxin electron carrier (RubA, S2)	**covered** as a rubredoxin carrier, but **candidate_uncertain** for the *alkane-specific* role	Gene present and a bona fide rubredoxin; alkane coupling unproven/over-propagated
Alkane 1-monooxygenase (S3)	**gap → not_expected_in_target_taxon**	No true AlkB; KT2440 is a non-alkanotroph; PP_3400 is a homonym
Whole UPA00191 module	**module_needs_revision** / not satisfiable	Terminal enzyme absent ⇒ pathway not satisfiable; boundary "rubredoxin = alkane degradation" is wrong for this organism

GO / annotation actions: 1. **Downgrade / requalify RubA's alkane annotations.** Recommend removing or qualifying `GO:0043448 (alkane catabolic process)` on RubA / PP_5371 for KT2440; retain generic `GO:0009055 (electron transfer activity)` and `GO:0005506 (iron ion binding)` as the experimentally-defensible core. 2. **Exclude PP_3400 from UPA00191.** Explicitly annotate it as the DNA-repair AlkB (K03919) to prevent name-based false satisfaction. 3. **Mark the module boundary as wrong for this organism.** The generic module presumes a downstream alkane hydroxylase that KT2440 lacks. Either (a) split the "rubredoxin electron transfer" carrier steps into an organism-agnostic redox module decoupled from alkane degradation, or (b) mark UPA00191 `not_expected_in_target_taxon` for KT2440. Consider a **NOT** qualifier or a request for a more accurate "rubredoxin-mediated electron transfer" grouping term. 4. **Consider a housekeeping-rubredoxin module/GO document.** RubA + AlkT (+PP_5371) constitute a general NAD(P)H:rubredoxin electron-transfer couple. A module scoped to "rubredoxin-mediated electron transfer / oxidative-stress redox" (not alkane catabolism) would represent the KT2440 biology more accurately.

7. Genes to Promote to Full Review

Gene	Locus	Why promote	Priority
rubA	PP_5315 (Q88C68)	Primary candidate; needs reclassification from "alkane-hydroxylase rubredoxin" to AlkG1-type/generic electron carrier; record the over-propagation caveat	**High** — promote to full `fetch-gene` review
PP_5371	PP_5371 (Q88C12)	Fused reductase–rubredoxin carrying identical over-propagated annotation; correct pathway assignment and clarify paralog relationship with AlkT	Medium
PP_3400	PP_3400 (Q88HF9)	Homonym "alkB"; fix to K03919 DNA-repair dioxygenase and explicitly exclude from the alkane module	Medium (documentation)
alkT	PP_5314 (Q88C69)	Confirm as the RubA reductase partner; annotation appears correct	Low

8. Mechanistic Model / Interpretation

The evidence converges on a clean, coherent picture. KT2440 possesses the **upstream electron-supply half** of the canonical alkane-oxidation relay but **not the downstream catalytic half**:

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  Present in KT2440 (covered)              Absent in KT2440 (gap)
 ┌───────────────────────────────────┐   ┌───────────────────────────────┐
 │ NADH → AlkT (PP_5314, FAD, K05297)│   │  Alkane 1-monooxygenase AlkB    │
 │         ↓                         │   │  (EC 1.14.15.3, membrane        │
 │      RubA (PP_5315, Fe, 55 aa)    │ ✗ │   non-heme diiron, ~400 aa)     │
 │   [AlkG1-type housekeeping Rd]    │   │  → NOT in proteome              │
 │         + PP_5371 (fused, K05297) │   │  PP_3400 "alkB" = DNA-repair    │
 └───────────────────────────────────┘  │ dioxygenase K03919 (homonym)   │
                                           └───────────────────────────────┘
          general redox / ppu00071
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Rubredoxin functional typing is the linchpin. Van Beilen et al. showed that Alk-rubredoxins divide into AlkG1 (conserved, housekeeping, non-complementing) and AlkG2/Rd2 (alkane-degrader-specific, complementing) classes, and that the alkane-functional donor is the **two-domain** AlkG. RubA's single ~55-aa domain, its 90% identity to a conserved *P. oleovorans* housekeeping rubredoxin, and its ~55% identity to true two-domain AlkG together place it firmly in the AlkG1 class. In a non-alkanotroph like KT2440, the parsimonious interpretation is that RubA + AlkT provide **general NAD(P)H:rubredoxin electron transfer** — a redox-housekeeping function (KEGG associates the reductases with fatty-acid degradation, ppu00071) — and that the UPA00191/alkane annotation was inherited by homology-based propagation from the well-studied GPo1 Alk system rather than reflecting KT2440 biology.

The practical consequence for module curation is that satisfiability cannot be inferred from the presence of rubredoxin + reductase alone. The module's biological meaning depends on the terminal hydroxylase, which is missing. The "alkB" homonym makes automated satisfiability especially treacherous here — a name-based match on PP_3400 would produce a spurious "complete pathway" call.

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9. Evidence Base

PMID	Title (abbrev.)	How it supports the findings
11872724	*Rubredoxins involved in alkane oxidation.*	**Central.** Establishes AlkG1 (housekeeping, non-complementing, conserved in non-degraders) vs AlkG2/Rd2 (alkane-functional, two-domain) classification; underpins F003 and F004 typing of RubA as AlkG1-housekeeping and its over-propagated alkane annotation.
34990970	*An alkane monooxygenase (AlkB) family...covalently bound electron transfer partners.*	Defines the canonical NADH → AlkT → rubredoxin (AlkG) → AlkB electron chain, situating RubA + AlkT as the reductase/carrier components (F001).
16958757	*Inactivation of the P. putida cytochrome o ubiquinol oxidase...*	Context on KT2440's respiratory chain and global regulation of "alkane assimilation" genes; indicates alkane-response regulation exists but does not establish a functional alkane hydroxylase.

Verbatim support snippets used in the findings: - van Beilen (PMID: 11872724): *"All alkane-degrading strains contain AlkG2-type Rds, which are able to replace the GPo1 Rd 2 in n-octane hydroxylation. Most strains also contain AlkG1-type Rds, which do not complement the deletion mutant but are highly conserved among gram-positive and gram-negative bacteria."* - van Beilen (PMID: 11872724): *"Complementation tests in an Escherichia coli recombinant containing all Pseudomonas putida GPo1 genes necessary for growth on alkanes except Rd 2 (AlkG) and sequence comparisons showed that the Alk-Rds can be divided in AlkG1- and AlkG2-type Rds."* - Williams et al. (PMID: 34990970): *"Electrons are typically supplied by NADH via a rubredoxin reductase (AlkT) to a rubredoxin (AlkG) to AlkB, although alternative electron transfer partners have been observed."*

Database anchors: UniProtKB Q88C68 (RubA/PP_5315), Q88C69 (AlkT/PP_5314), Q88C12 (PP_5371), Q88HF9 (PP_3400); KEGG ppu:PP_5314/PP_5371 → K05297 (rubredoxin–NAD^+ reductase, EC 1.18.1.1), ppu:PP_3400 → K03919 (DNA oxidative demethylase AlkB); UniPathway UPA00191; comparator sequences A0A061CWA0 (*P. oleovorans* single-domain rubredoxin), P00272 (GPo1 AlkG/Rubredoxin-2), Q9WWW4 (*P. putida* AlkG).

Evidence provenance summary: - **Direct genomic (target-specific), strong:** existence and family of RubA/AlkT/PP_5371 (UniProt/KEGG, KT2440-specific); the "alkB" homonym identity of PP_3400. - **Experimental but from related organisms, transferred by typing:** AlkG1/AlkG2 complementation biology (van Beilen, GPo1/*P. oleovorans*). Strong for classifying RubA as AlkG1 and for concluding KT2440 lacks the OCT-plasmid Alk system. - **Inferred from databases/homology, weak for KT2440:** RubA's alkane-specific electron-transfer role (UPA00191, GO:0043448), transferred from GPo1 and undermined by the absence of a terminal AlkB.

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10. Limitations and Knowledge Gaps

1. **No direct KT2440 experiment defines RubA's physiological partner.** The conclusion that RubA is a general redox carrier rather than an alkane-hydroxylase donor is a strong inference (sequence class + absence of AlkB), not a direct enzymatic measurement in KT2440.
2. **Sequence typing used anchored pairwise identities, not a full phylogeny.** A formal maximum-likelihood tree of AlkG1 vs AlkG2 rubredoxins with KT2440 RubA included would harden the classification, though the single-domain architecture already discriminates cleanly.

3. **Proteome-wide search for a cryptic alkane hydroxylase was targeted, not exhaustive.** KT2440 could in principle oxidize hydrocarbons via unrelated enzyme families (e.g., cytochrome P450s, AlmA-type flavin monooxygenases, LadA); these were not systematically enumerated. However, none would satisfy the *rubredoxin-AlkB* UPA00191 model.
4. **Physiological role of the two reductases (PP_5314, PP_5371) is unresolved.** Whether they are functionally redundant, differentially regulated, or serve distinct acceptors is unknown.
5. **Plausible alternative role is inferential.** In diverse bacteria, rubredoxin/rubredoxin-reductase couples act as general NAD(P)H electron carriers and in oxidative/nitrosative-stress defense (e.g., rubredoxin:oxygen oxidoreductase systems; Figueiredo et al. 2013, PMID: 23564166, in the anaerobe *Desulfovibrio* — so species transfer to the aerobe KT2440 is **weak/uncertain**). This is the most plausible physiological function of the KT2440 RubA+AlkT couple in the absence of an alkane hydroxylase, but there is no direct KT2440 experimental evidence.
6. **UniPathway/GO term structure.** Whether UPA00191 should be formally split (carrier step vs terminal hydroxylase) is a database-modeling question beyond this organism-level review.

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11. Proposed Follow-up Experiments / Actions

Curation actions (immediate, no bench work): 1. Requalify Q88C68 (RubA): remove/flag `GO:0043448` and the UPA00191 alkane-catabolic assignments as over-propagated IEA; retain `GO:0009055 electron transfer activity` and `GO:0005506 iron ion binding`. 2. Annotate PP_3400 explicitly as DNA-repair AlkB (K03919) and exclude it from UPA00191 satisfiability logic. 3. Set UPA00191 for KT2440 as **not satisfiable / not_expected_in_target_taxon** with the terminal AlkB step marked **gap**; add a note that the electron-relay genes are present but the pathway lacks its hydroxylase. 4. Promote `rubA` (PP_5315) to full `fetch-gene` review; secondary review of PP_5371 and PP_3400.

Computational follow-ups: 5. Build a maximum-likelihood phylogeny of AlkG1/AlkG2 rubredoxins including RubA to formally confirm AlkG1 placement. 6. Systematically scan UP000000556 for any alternative alkane-oxidizing families (P450, AlmA, LadA) to document, definitively, the absence of an alkane hydroxylase — and record that even if found, none would satisfy the rubredoxin-AlkB UPA00191 model.

Experimental resolution of open questions: 7. Growth/activity assays: confirm KT2440 cannot use medium- to long-chain n-alkanes as sole carbon source (expected negative), consistent with the absence of a functional Alk pathway. 8. In vitro reconstitution: assemble AlkT/PP_5371 + RubA and test NAD(P)H:rubredoxin electron transfer and candidate physiological acceptors (fatty-acid desaturase/oxidase systems, oxidative-stress partners). 9. Genetics/transcriptomics: phenotype of ΔrubA under oxidative/nitrosative stress vs hydrocarbon exposure; test co-regulation of PP_5314–PP_5315 to identify the true redox partner.

Consensus Verdict

The rubredoxin electron-transfer *components* of UPA00191 are genuinely encoded in *P. putida* KT2440 — RubA (PP_5315), its adjacent FAD rubredoxin–NAD(+) reductase AlkT (PP_5314), and a fused reductase-rubredoxin PP_5371 — so the electron-carrier step is **covered**. But the pathway's terminal alkane 1-monooxygenase (AlkB) is **absent**: the only "alkB"-named gene (PP_3400) is the DNA-repair dioxygenase AlkB (K03919), a homonym. UPA00191 alkane degradation is therefore **not satisfiable** in this non-alkanotroph, and RubA — a conserved single-domain AlkG1-type housekeeping rubredoxin (90% identical to a conserved *P. oleovorans* rubredoxin, only ~55% to two-domain alkane-transfer AlkG/Rd2) — carries a **likely over-propagated** alkane-hydroxylase annotation; it most plausibly functions as a general NAD(P)H:rubredoxin electron carrier.
